CHAPTER 3
the binomial model

The aim of this Chapter...

...is to describe the simplest model for asset prices that can be, and is, used for pricing derivatives, to introduce the two fundamental building blocks of quantitative finance and to explain a very important (but very counter-intuitive) financial concept. By the end of this chapter you will be able to write a program for pricing basic derivatives. I hope you won't be too confused by the important-but-counter-intuitive financial concept... even if you are, we'll come back to it many more times.

In this Chapter...

- a simple model for an asset price random walk
- delta hedging
- no arbitrage
- the basics of the binomial method for valuing options
- risk neutrality

3.1 INTRODUCTION

In this chapter I'm going to present a very simple and popular model for the random behavior of an asset, for the moment think 'equity.' This simple model will allow us to start valuing options. Undoubtedly, one of the reasons for the popularity of this model is that it can be implemented without any higher mathematics (such as differential calculus). This is a positive point, however the downside is that it is harder to attain greater levels of sophistication or numerical analysis in this setting.

Later we'll be seeing a more sophisticated model, but the ideas we first encounter in this chapter will be seen over and over again. These are the fundamental concepts of hedging and no arbitrage.

The binomial model is very important because it shows that you don't need a simple formula for everything. Indeed, it is extremely important to have a way of valuing options that only relies on a simple model and fast, accurate numerical methods. Often in real life a contract may contain features that make analytic solution very hard or impossible. Some of these features may be just a minor modification to some other, easily-priced, contract but even minor changes to a contract can have important effects on the value and especially on the method of solution. The classic example is of the American put. Early exercise may seem to be a small change to a contract but the difference between the values of a European and an American put can be large and certainly there is no simple closed-form solution for the American option and its value must be found numerically.

Time Out...

Simplicity itself

The math in this chapter is all very straightforward, addition, subtraction, multiplication and, occasionally, division.

Before I describe this model I want to stress that the binomial model may be thought of as being either a genuine *model* for the behavior of equities, or, alternatively, as a *numerical method* for the solution of the Black–Scholes equation.[1] Or it can be thought of as a *teaching aid* to explain delta hedging, risk elimination and risk-neutral valuation.

Having said what is good about the binomial I would like to say what, to my mind, is bad about it.

First, as a model of stock price behavior it is poor. The binomial model says that the stock can either go up by a known amount or down by a known amount, there are but two possible stock prices 'tomorrow.' This is clearly unrealistic. This is important because

[1] In this case, it is very similar to an explicit finite-difference method. We'll see the famous Black–Scholes equation and finite-difference methods later.

from this model follow certain results that hinge entirely on there only being two prices for the stock tomorrow. Introduce a third state and the results collapse.

Second, as a numerical scheme it is prehistoric compared with modern numerical methods. We go into these numerical methods in some detail in this book but several volumes could be written on sophisticated numerical methods alone. I would advise the reader to study the binomial model for the intuition it gives, but do not rely on it for numerical calculations.

But as a teaching aid for explaining difficult concepts, the binomial model is fantastic. Indeed, it is said that the binomial model even helps MBA students understand options. However, I don't believe in such dumbing down. I don't think that quantitative finance should be dumbed down, just like I don't believe that brain surgery should be dumbed down.

My advice is that once you have become comfortable with the ideas that come out of this chapter you should relegate the binomial method to the back of your mind.

3.2 EQUITIES CAN GO DOWN AS WELL AS UP

The most 'accessible' approach to option pricing is the **binomial model**. This requires only basic arithmetic and no complicated stochastic calculus. In this model we will see the ideas of hedging and no arbitrage used. The end result is a simple algorithm for determining the correct value for an option.

We are going to examine a very simple model for the behavior of a stock, and based on this model see how to value options.

- We will have a stock, and a call option on that stock expiring tomorrow.
- The stock can either rise or fall by a known amount between today and tomorrow.
- Interest rates are zero.

Figure 3.1 gives an example. The stock is currently worth \$100 and can rise to \$101 or fall to \$99 between today and tomorrow.

Which of the two prices is realized tomorrow is completely random. There is a certain probability of the stock rising and one minus that probability of the stock falling. In this example the probability of a rise to 101 is 0.6, so that the probability of falling to 99 is 0.4. See Figure 3.2.

Now let's introduce the call option on the stock. This call option has a strike of \$100 and expires tomorrow.

If the stock price rises to 101, what will then be the option's payoff? See Figure 3.3. It is just $101 - 100 = 1$.

And if the stock falls to 99 tomorrow, what is then the payoff? See Figure 3.4. The answer is zero, the option has expired out of the money.

If the stock rises the option is worth 1, and if it falls it is worth 0. There is a 0.6 probability of getting 1 and a 0.4 probability of getting zero. Interest rates are zero...

What is the option worth today?

No, the answer is *not* 0.6. If that is what you thought, based on calculating simple expectations, then I successfully 'led you up the garden path' to the wrong answer.

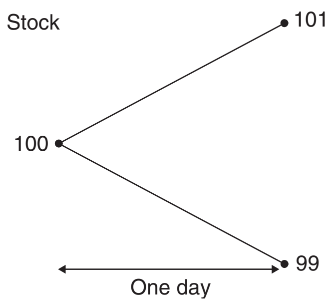


Figure 3.1 The stock can rise or fall over the next day, only two future prices are possible.

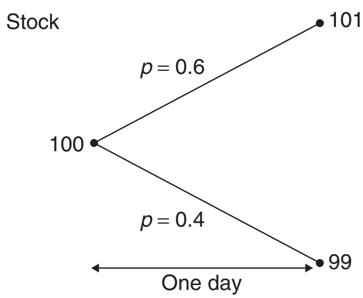


Figure 3.2 Probabilities associated with the future stock prices.

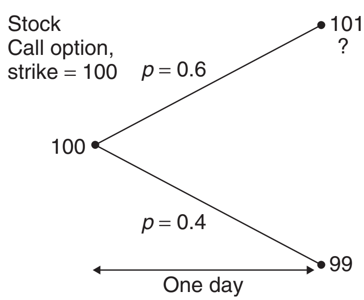


Figure 3.3 What is the option payoff if the stock rises?

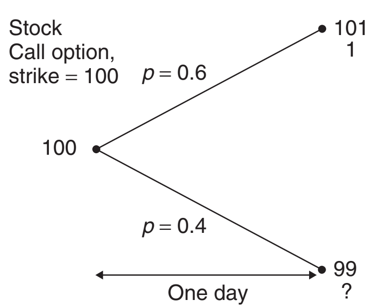


Figure 3.4 What is the option payoff if the stock falls?

3.3 THE OPTION VALUE

The correct answer is. . .

$$\frac{1}{2}.$$

Why?

To see how this can be the only correct answer we must first construct a *portfolio* consisting of one option and short $\frac{1}{2}$ of the underlying stock. This portfolio is shown in Figure 3.7.

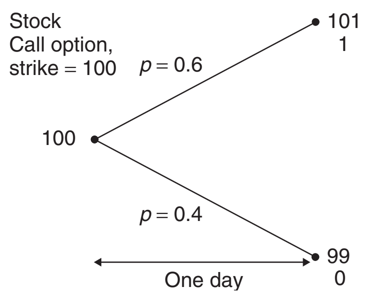


Figure 3.5 Now we know the option values in both 'states of the world.'

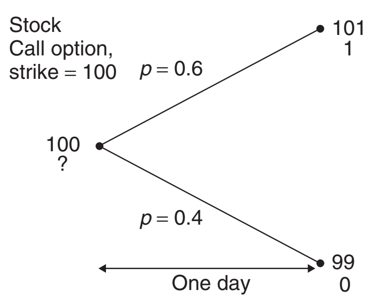


Figure 3.6 What is the option worth today?

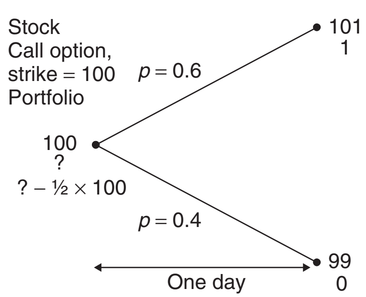


Figure 3.7 Long one option, short half of the stock.

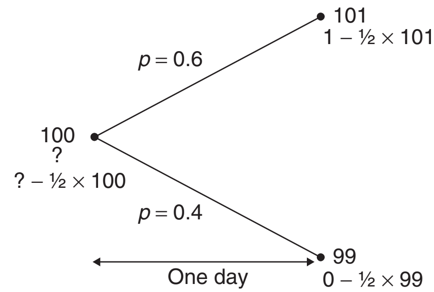


Figure 3.8 The portfolio values at expiration.

If the stock rises to 101 then this portfolio is worth

$$1 - \tfrac{1}{2} \times 101;$$

the one being from the option payoff and the $-\frac{1}{2} \times 101$ being from a short (–) position ($\frac{1}{2}$) in the stock (now worth 101).

If the stock falls to 99 then this portfolio is worth

$$0 - \tfrac{1}{2} \times 99;$$

the zero being from the option payoff and the $-\frac{1}{2} \times 99$ being from a short (–) position ($\frac{1}{2}$) in the stock (now worth 99). See Figure 3.8.

In either case, tomorrow, at expiration, the portfolio takes the value

$$-\frac{99}{2}$$

and that is regardless of whether the stock rises or falls.

We have constructed a perfectly risk-free portfolio.

If the portfolio is worth $-99/2$ tomorrow, and interest rates are zero, how much is this portfolio worth today?

It must also be worth $-99/2$ today.

This is an example of **no arbitrage**: There are two ways to ensure that we have $-99/2$ tomorrow.

1. Buy one option and sell one half of the stock.
2. Put the money under the mattress.

Both of these 'portfolios' must be worth the same today. Therefore, using '?' as in the figure to represent the unknown option value

$$? - \tfrac{1}{2} \times 100 = \text{the option value} - \tfrac{1}{2} \times 100 = -\tfrac{1}{2} \times 99$$

and so

$$? = \text{the option value} = \frac{1}{2}.$$

3.4 WHICH PART OF OUR 'MODEL' DIDN'T WE NEED?

The value of an option does not depend on the probability of the stock rising or falling. This is equivalent to saying that the stock growth rate is irrelevant for option pricing. This is because we have **hedged** the option with the stock. See Figure 3.9. We do not care whether the stock rises or falls. See Figure 3.10.

We *do* care about the stock price range, however. The stock volatility is very important in the valuation of options.

Three questions follow from the above simple argument:

- Why should this 'theoretical price' be the 'market price'?
- How did I know to sell $\frac{1}{2}$ of the stock for hedging?
- How does this change if interest rates are non-zero?

3.5 WHY SHOULD THIS 'THEORETICAL PRICE' BE THE 'MARKET PRICE'?

This one is simple. Because if the theoretical price and the market price are not the same, then there is risk-free money to be made. If the option costs less than 0.5 simply buy it and hedge to make a profit. If it is worth more than 0.5 in the market then sell it and hedge, and make a guaranteed profit.

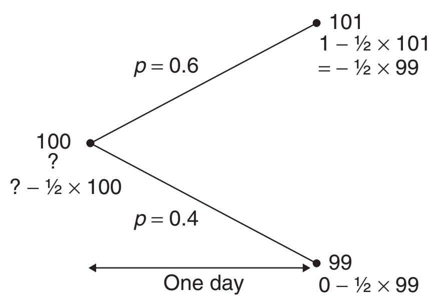


Figure 3.9 The portfolio is 'hedged.'

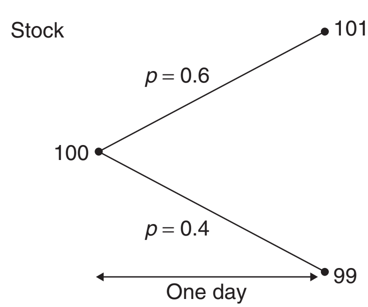


Figure 3.10 Which parameter(s) didn't we need?

It's not quite this simple because it is possible for arbitrage opportunities to exist, and to exist for a long time. We also really need there to be some practical mechanism for arbitrage to be 'removed.' In practice this means we really need there to be a couple of agents perhaps undercutting each other in such a way that the arb opportunity disappears: A sells the option for 0.55, gets all the business and makes a guaranteed 0.05 profit. Along comes B who sells the option for just 0.53, now he takes away all the business from A, who responds by dropping his price to 0.52, etc. So really, supply and demand should act to make the option price converge to the 0.5.

3.5.1 The role of expectations

The expected payoff is definitely 0.6 for this option. It's just that this has nothing to do with the option's value. Let's take a quick look at the role of this expectation.

Would anyone pay 0.6 or more for the option? No, unless they were **risk seeking**.

Would anyone pay 0.55? Perhaps, if they liked the idea of an expected return of

$$\frac{0.6 - 0.55}{0.55} \approx 9\%.$$

The person writing the option would be very pleased with the guaranteed profit of 0.05.

3.6 HOW DID I KNOW TO SELL $\frac{1}{2}$ OF THE STOCK FOR HEDGING?

Introduce a symbol. Use Δ to denote the quantity of stock that must be sold for hedging. We start off with one option, $-\Delta$ of the stock, giving a portfolio value of

$$? - \Delta \times 100.$$

Tomorrow the portfolio is worth

$$1 - \Delta \times 101$$

if the stock rises, or

$$0 - \Delta \times 99$$

if it falls.

The key step is the next one; make these two equal to each other:

$$1 - \Delta \times 101 = 0 - \Delta \times 99.$$

Therefore

$$\Delta(101 - 99) = 1$$

$$\Delta = 0.5.$$

Another example: Stock price is 100, can rise to 103 or fall to 98. Value a call option with a strike price of 100. Interest rates are zero.

Again use Δ to denote the quantity of stock that must be sold for hedging.

The portfolio value is

$$? - \Delta \times 100.$$

Tomorrow the portfolio is worth either

$$3 - \Delta 103$$

or

$$0 - \Delta 98.$$

So we must make

$$3 - \Delta 103 = 0 - \Delta 98.$$

That is,

$$\Delta = \frac{3-0}{103-98} = \frac{3}{5} = 0.6.$$

The portfolio value tomorrow is then

$$-0.6 \times 98.$$

With zero interest rate, the portfolio value today must equal the risk-free portfolio value tomorrow:

$$? - 0.6 \times 100 = -0.6 \times 98.$$

Therefore the option value is 1.2.

3.6.1 The general formula for Δ

Delta hedging means choosing Δ such that the portfolio value does not depend on the direction of the stock.

When we generalize this (using symbols instead of numbers later on) we will find that

$$\Delta = \frac{\text{Range of option payoffs}}{\text{Range of stock prices}}.$$

We can think of Δ as the sensitivity of the option to changes in the stock.

3.7 HOW DOES THIS CHANGE IF INTEREST RATES ARE NON-ZERO?

Simple. We delta hedge as before to construct a risk-free portfolio. (And we use exactly the same delta.) Then we present value that back in time, by multiplying by a discount factor.

Example: Same as first example, but now $r = 0.1$.

The discount factor for going back one day is

$$\frac{1}{1+0.1/252} = 0.9996.$$

The portfolio value today must be the *present value* of the portfolio value tomorrow

$$? - 0.5 \times 100 = -0.5 \times 99 \times 0.9996.$$

So that

$$? = 0.51963.$$

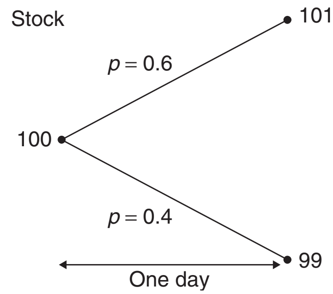


Figure 3.11 What is the expected stock price?

3.8 IS THE STOCK ITSELF CORRECTLY PRICED?

Earlier, I tried to trick you into pricing the option by looking at the expected payoff. Suppose, for the sake of argument, that I had been successful in this. I would then have asked you what was the expected stock price tomorrow; forget the option.

The expected stock value tomorrow is

$$0.6 \times 101 + 0.4 \times 99 = 100.2.$$

In an expectation's sense, the stock itself seems incorrectly priced. Shouldn't it be valued at 100.2 today? Well, we already kind of know that expectations aren't the way to price options, so this is also not the way to price stock. But we can go further than that, and make some positive statements.

We ought to pay less than the future expected value because the stock is risky. We want a positive expected return to compensate for the risk. This is an idea we will be seeing in detail later on, in Chapter 21 on portfolio management.

We can plot the stock (and all investments) on a risk/return diagram, see Figure 3.12. Risk is measured by standard deviation and return is the expected return. The figure shows two investments, the stock and, at the origin, the bank investment. The bank investment has zero risk, and in our first example above, has zero expected return.

We will see in Chapter 21 that we can get to other places in the risk/return space by dividing our money between several investments. In the present case, if we put half our

Figure 3.12 Risk and return for the stock and the risk-free investment.

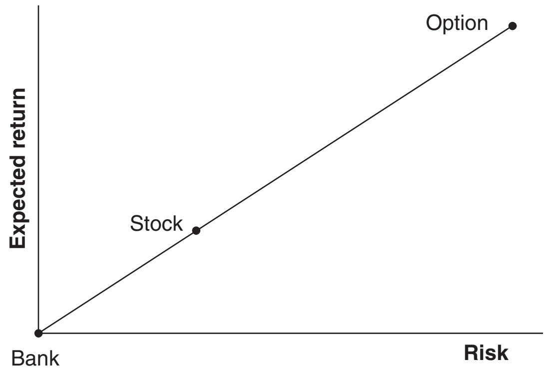


Figure 3.13 Now we have three investments, including the option.

money in the bank and half in the stock we will find ourselves with an investment that is exactly half way between the two dots in the figure. We can get to any point on the straight line between the risk-free dot and the stock dot by splitting our money between these two, we can even get to any place on the extrapolated straight line in Figure 3.13 by borrowing money at the risk-free rate to invest in the stock.

3.9 COMPLETE MARKETS

The option also has an expected return and a risk. In our example the expected return and the risk for the option are both much, much greater than for the stock. We can plot the option on the same risk/return diagram. Where do you think it might be? Above the extrapolated line, on it, or below it?

It turns out that the option lies *on* the straight line, see Figure 3.13. This means that we can 'replicate' an option's risk and return characteristic with stock and the risk-free investment. Option payoffs can be **replicated** by stocks and cash. Any two points on the straight lines can be used to get us to any other point. So, we can get a risk-free investment using the option and the stock, and this is hedging. And, the stock can be replicated by cash and the option.

A conclusion of this analysis is that options are redundant in this 'world,' i.e. in this model. We say that **markets are complete**. The practical implication of complete markets is that options are hedgeable and can therefore be priced without any need to know probabilities. We can hedge an option with stock to 'replicate' a risk-free investment, Figure 3.14, and we can replicate an option using stock and a risk-free investment, Figures 3.15 and 3.16. We could also, of course, replicate stock with an option and risk-free investment.

3.10 THE REAL AND RISK-NEUTRAL WORLDS

In our world, the **real world**, we have used our statistical skills to estimate the future possible stock prices (99 and 101) and the probabilities of reaching them (0.4 and 0.6).

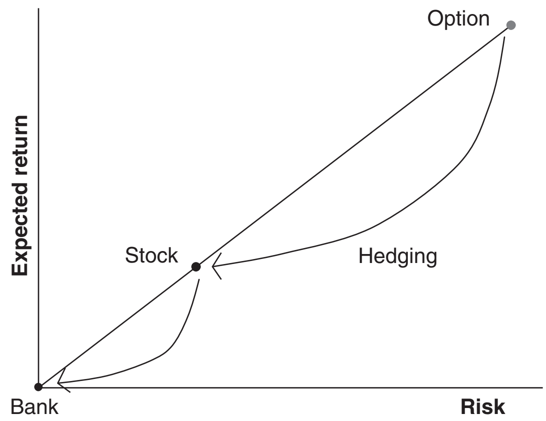


Figure 3.14 Hedging.

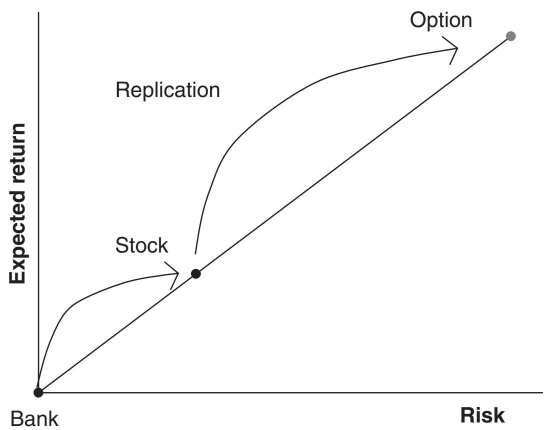


Figure 3.15 Replication.

Some properties of the real world:

- We know all about delta hedging and risk elimination.
- We are very sensitive to risk, and expect greater return for taking risk.
- It turns out that only the two stock prices matter for option pricing, not the probabilities.

People often refer to the **risk-neutral world** in which people don't care about risk. The risk-neutral world has the following characteristics:

- We don't care about risk, and don't expect any extra return for taking unnecessary risk.
- We don't ever need statistics for estimating probabilities of events happening.
- We believe that everything is priced using simple expectations.

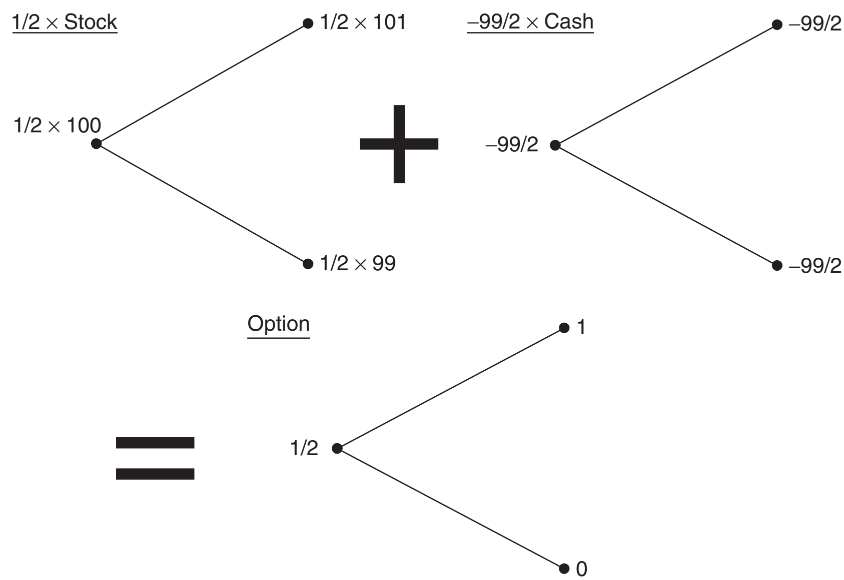


Figure 3.16 $1/2 \times$ Stock $- 99/2 \times$ Cash $=$ Option.

Imagine yourself in the risk-neutral world, looking at the stock price model. Suppose all you know is that the stock is currently worth \$100 and could rise to \$101 or fall to \$99.

If the stock is correctly priced today, using simple expectations, what would you deduce to be the probabilities of the stock price rising or falling?

The symmetry makes the answer to this rather obvious. If the stock is correctly priced using expectations then the probabilities ought to be 50% chance of a rise and 50% chance of a fall. The calculation we have just performed goes as follows. . .

On the risk-neutral planet they calculate **risk-neutral probabilities** p' from the equation

$$p' \times 101 + (1 - p') \times 99 = 100.$$

From which $p' = 0.5$.

Do not think that this p' is in any sense real. No, the real probabilities are still 60% and 40%. This calculation assumes something that is fundamentally wrong, that simple expectations are used for pricing. No allowance has been made for risk.

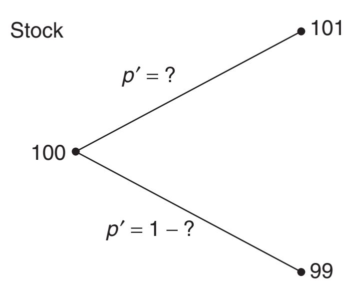


Figure 3.17 What is the probability of the stock rising?

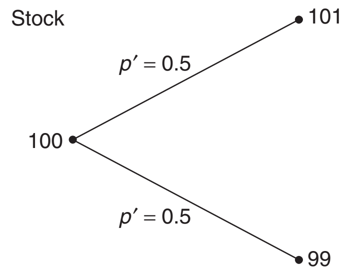


Figure 3.18 Risk-neutral probabilities.

Never mind, let's stay with this risk-neutral world and see what they think the option value is. We won't tell them yet that the calculation they have just done is 'wrong.'

How would they then value the call option? Since they reckon the probabilities to be 50–50 and they use simple expectations to calculate values with no regard to risk then they would price the option using the expected payoff with their probabilities, i.e.

$$0.5 \times 1 + 0.5 \times 0 = 0.5.$$

See Figure 3.19. This is called the **risk-neutral expectation**.

Damn and blast! They have found the correct answer for the wrong reasons! To put it in a nutshell, they have twice used their basic assumption of pricing via simple expectations to get to the correct answer. Two wrongs in this case do make a right. First of all they calculate a probability from a price, and then a price from a probability. The two 'errors' were in opposite directions and canceled each other out.

And this technique will always work.

In the risk-neutral world they have exactly the same price for the option (but for different reasons).

3.10.1 Non-zero interest rates

When interest rates are non-zero we must perform exactly the same operations, but whenever we equate values at different times we must allow for present valuing.

With $r = 0.1$ we calculate the risk-neutral probabilities from

$$0.9996 \times \left(p' \times 101 + (1 - p') \times 99\right) = 100.$$

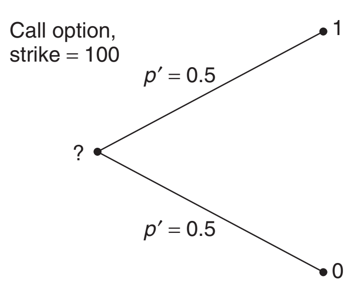


Figure 3.19 Pricing the option.

So

$$p' = 0.51984.$$

The expected option payoff is now

$$0.51984 \times 1 + (1 - 0.51984) \times 0 = 0.51984.$$

And the present value of this is

$$0.9996 \times 0.51984 = 0.51963.$$

And this must be the option value. (It is the same as we derived the 'other' way.)

Risk-neutral pricing is a very powerful technique, and we will be seeing a lot more of it. Just remember one thing for the moment, that the risk-neutral probability p' that we have just calculated (the 0.5 in the first example) is not real, it does not exist, it is a mathematical construct. The real probability of the stock price was always in our example 0.6, it's just that this never was used in our calculations.

3.11 AND NOW USING SYMBOLS

Let's generalize, so that instead of 100, 101, 99, 0.6, etc., we use symbols for everything. In the binomial model we assume that the asset, which initially has the value S, can, during a time step δt, either

- rise to a value $u \times S$ or
- fall to a value $v \times S$,

with $0 < v < 1 < u$.

- The probability of a rise is p and so the probability of a fall is $1 - p$.

Note: By *multiplying* the asset price by constants rather than *adding* constants, we will later be able to build up a whole tree of prices. (This will be a discrete-time version of what you will soon come to know as a lognormal random walk.)

- The three constants u, v and p are chosen to give the binomial walk the same characteristics as the asset we are modeling.

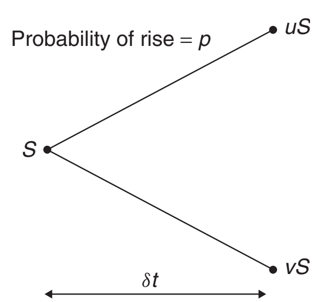


Figure 3.20 The model, using symbols.

Now remember that I said that I don't want you using the binomial model, other than to gain intuition? Well, to help you in that I want to wean you off the model starting now. To that end I am going to use notation that will be important from Chapter 4 onwards, and is notation we all use in proper quant finance. Instead of using, u, v and p we are going to write everything in terms of the mean and the standard deviation of the random walk. That involves writing u, v and p in terms of what are known as the drift and the volatility. The drift is the average rate at which the asset rises and the volatility is a measure of its randomness.

In order to do this I have to introduce new symbols to represent the drift and volatility of the asset: μ the drift of the asset and σ the volatility. And we still have a time step δt over which the asset move takes place.

I'm going to give some expressions now for u, v and p and then explain where they come from:

$$u = 1 + \sigma\sqrt{\delta t},$$

$$v = 1 - \sigma\sqrt{\delta t}$$

and

$$p = \frac{1}{2} + \frac{\mu\sqrt{\delta t}}{2\sigma}. \tag{3.1}$$

To see what these mean let's look at the average change in asset price during the time step and the standard deviation.

3.11.1 Average asset change

The expected asset price after one time step is

$$puS + (1-p)vS = \left(\frac{1}{2} + \frac{\mu\sqrt{\delta t}}{2\sigma}\right)\left(1 + \sigma\sqrt{\delta t}\right)S + \left(\frac{1}{2} - \frac{\mu\sqrt{\delta t}}{2\sigma}\right)\left(1 - \sigma\sqrt{\delta t}\right)S = (1 + \mu\,\delta t)S.$$

So the expected change in the asset is $\mu S\,\delta t$.

- The expected **return** is $\mu\,\delta t$.

3.11.2 Standard deviation of asset price change

The variance of change in asset price is

$$\begin{aligned} & S^2\left(p(u - 1 - \mu\,\delta t)^2 + (1-p)(v - 1 - \mu\,\delta t)^2\right) \\ &= S^2\left(\left(\frac{1}{2} + \frac{\mu\sqrt{\delta t}}{2\sigma}\right)\left(\sigma\sqrt{\delta t} - \mu\,\delta t\right)^2 + \left(\frac{1}{2} - \frac{\mu\sqrt{\delta t}}{2\sigma}\right)\left(\sigma\sqrt{\delta t} + \mu\,\delta t\right)^2\right) \\ &= S^2(\sigma^2\delta t - \mu^2\delta t^2). \end{aligned}$$

The standard deviation of asset changes is (approximately) $S\sigma\sqrt{\delta t}$.

- The standard deviation of returns is (approximately) $\sigma\sqrt{\delta t}$.

3.12 AN EQUATION FOR THE VALUE OF AN OPTION

Suppose that we know the value of the option at the time $t + \delta t$. For example, this time may be the expiration of the option, say.

Now construct a portfolio at time t consisting of one option and a short position in a quantity Δ of the underlying. At time t this portfolio has value

$$\Pi = V - \Delta S,$$

where the option value V is for the moment unknown. You'll recognize this as exactly what we did before, but now we're using symbols instead of numbers.

At time $t + \delta t$ the option takes one of two values, depending on whether the asset rises or falls

$$V^+ \quad \text{or} \quad V^-.$$

At the same time the portfolio becomes either

$$V^+ - \Delta uS \quad \text{or} \quad V^- - \Delta vS.$$

Since we know V^+, V^-, u, v and S the values of both of these expressions are just linear functions of Δ.

3.12.1 Hedging

Having the freedom to choose Δ, we can make the value of this portfolio the same whether the asset rises or falls. This is ensured if we make

$$V^+ - \Delta uS = V^- - \Delta vS.$$

This means that we should choose

$$\Delta = \frac{V^+ - V^-}{(u - v)S} \tag{3.2}$$

for hedging. This is just the range of option prices divided by the range of asset prices.

The portfolio value is then

$$V^+ - \Delta uS = V^+ - \frac{u(V^+ - V^-)}{(u - v)}$$

if the stock rises or

$$V^- - \Delta vS = V^- - \frac{v(V^+ - V^-)}{(u - v)}$$

if it falls.

And, of course, these two expressions are the same.

Let's denote this portfolio value by

$$\Pi + \delta\Pi.$$

This just means the original portfolio value plus the change in value.

3.12.2 No arbitrage

Since the value of the portfolio has been guaranteed, we can say that its value must coincide with the value of the original portfolio plus any interest earned at the risk-free rate; this is the no-arbitrage argument.

Thus

$$\delta\Pi = r\Pi\,\delta t.$$

Putting everything together we get

$$\Pi + \delta\Pi = \Pi + r\Pi\,\delta t = \Pi(1 + r\,\delta t) = V^+ - \frac{u(V^+ - V^-)}{(u - v)}$$

with

$$\Pi = V - \Delta S = V - \frac{V^+ - V^-}{(u - v)S}S = V - \frac{V^+ - V^-}{(u - v)}$$

And the end result is

$$(1 + r\,\delta t)\left(V - \frac{V^+ - V^-}{(u - v)}\right) = V^- - \frac{v(V^+ - V^-)}{(u - v)}.$$

On a spreadsheet

Rearranging as an equation for V we get

$$(1 + r\,\delta t)V = (1 + r\,\delta t)\frac{V^+ - V^-}{u - v} + \frac{uV^- - vV^+}{(u - v)}.$$

This is an equation for V given V^+, and V^-, the option values at the next time step, and the parameters u and v describing the random walk of the asset.

But it can be written more elegantly than this.

This equation can also be written as

$$V = \frac{1}{1 + r\,\delta t}\left(p'V^+ + (1 - p')V^-\right), \tag{3.3}$$

where

$$p' = \frac{1}{2} + \frac{r\sqrt{\delta t}}{2\sigma}. \tag{3.4}$$

The right-hand side of Equation (3.3) is just like a discounted expectation; the terms inside the parentheses are the sum of probabilities multiplied by events.

If only the expression contained p, the real probability of a stock rise, then this expression would be the expected value at the next time step.

We see that the probability of a rise or fall is irrelevant as far as option pricing is concerned since p did not appear in Equation (3.3). But what if we interpret p' as a probability? Then we could 'say' that the option price is the present value of an expectation. But not the real expectation.

We are back with risk-neutral expectations again.

Let's compare the expression for p' with the expression for the actual probability p:

$$p' = \frac{1}{2} + \frac{r\sqrt{\delta t}}{2\sigma}$$

but

$$p = \frac{1}{2} + \frac{\mu\sqrt{\delta t}}{2\sigma}.$$

The two expressions differ in that where one has the interest rate r the other has the drift μ, but are otherwise the same. Strange.

- We call p' the **risk-neutral probability**. It's like the real probability, but the real probability if the drift rate were r instead of μ.

Observe that the risk-free interest rate plays two roles in option valuation. It's used once for discounting to give present value, and it's used as the drift rate in the risk-neutral asset price random walk.

3.13 WHERE DID THE PROBABILITY *p* GO?

What happened to the probability p and the drift rate μ?

Interpreting p' as a probability, (3.3) is the statement that

- the option value at any time is the present value of the risk-neutral expected value at any later time.

In reading books or research papers on mathematical finance you will often encounter the expression 'risk-neutral' this or that, including the expression risk-neutral probability. You can think of an option value as being the present value of an expectation, only it's not the real expectation.

Don't worry we'll come back to this several more times until you get the hang of it.

3.14 COUNTER-INTUITIVE?

- Two stocks A and B.
- Both have the same value, same volatility and are denominated in the same currency.
- Both have call options with the same strike and expiration.
- Stock A is doubling in value every year, stock B is halving.

Therefore both call options have the same value. But which would you buy? That one stock is doubling and the other halving is irrelevant. That option prices don't depend on the direction that the stock is going can be difficult to accept initially.

3.15 THE BINOMIAL TREE

The binomial model, just introduced, allows the stock to move up or down a prescribed amount over the next time step. If the stock starts out with value S then it will take either the value uS or vS after the next time step. We can extend the random walk to the next time step. After two time steps the asset will be at either u^2S, if there were two up moves, uvS, if an up was followed by a down or vice versa, or v^2S, if there were two consecutive down moves. After three time steps the asset can be at u^3S, u^2vS, etc. One can imagine extending this random walk out all the way until expiry. The resulting structure looks like Figure 3.21 where the nodes represent the values taken by the asset. This structure is called the **binomial tree**. Observe how the tree bends due to the geometric nature of the asset growth. Often this tree is drawn as in Figure 3.22 because it is easier to draw, but this doesn't quite capture the correct structure.

The top and bottom branches of the tree at expiry can only be reached by one path each, either all up or all down moves. Whereas there will be several paths possible for each of the intermediate values at expiry. Therefore the intermediate values are more likely to be reached than the end values if one were doing a simulation. The binomial tree therefore contains within it an approximation to the probability density function for the lognormal random walk.

3.16 THE ASSET PRICE DISTRIBUTION

The probability of reaching a particular node in the binomial tree depends on the number of distinct paths to that node and the probabilities of the up and down moves. Since up and down moves are approximately equally likely and since there are more paths to the interior prices than to the two extremes we will find that the probability distribution of future prices is roughly bell shaped. In Figure 3.23 is shown the number of paths to each node after four time steps and the probability of getting to each. In Figure 3.24 this is interpreted as probability density functions at a sequence of times.

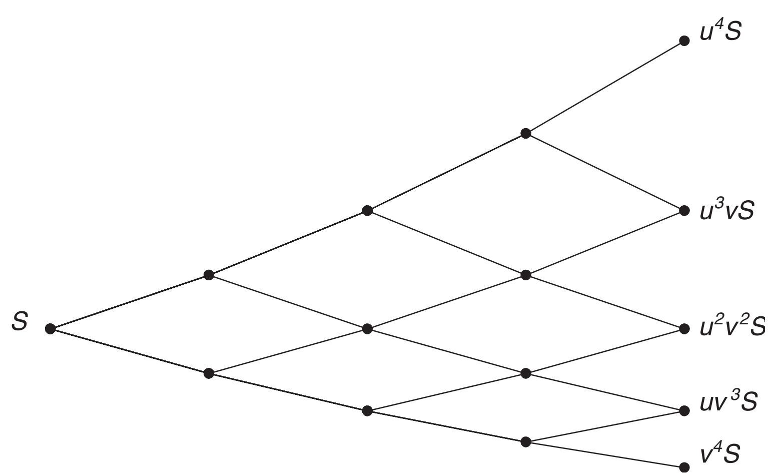


Figure 3.21 The binomial tree.

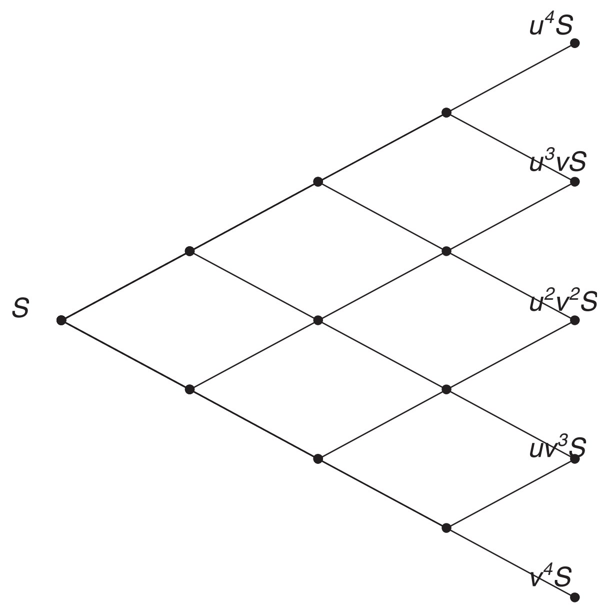


Figure 3.22 The binomial tree: a schematic version.

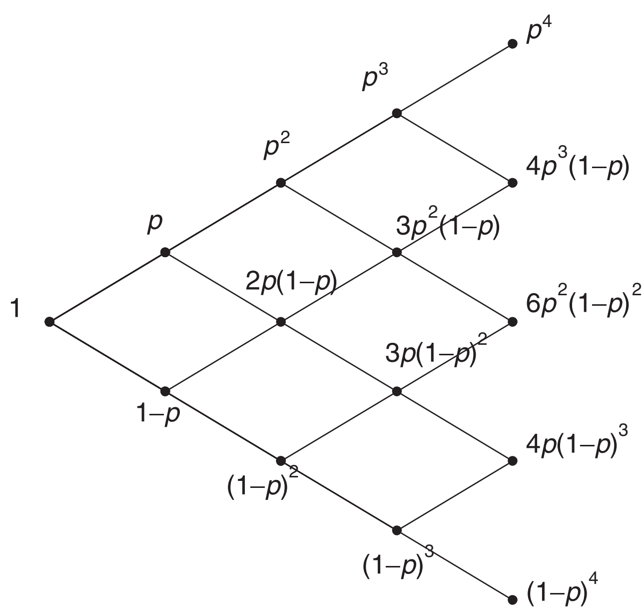


Figure 3.23 Counting paths.

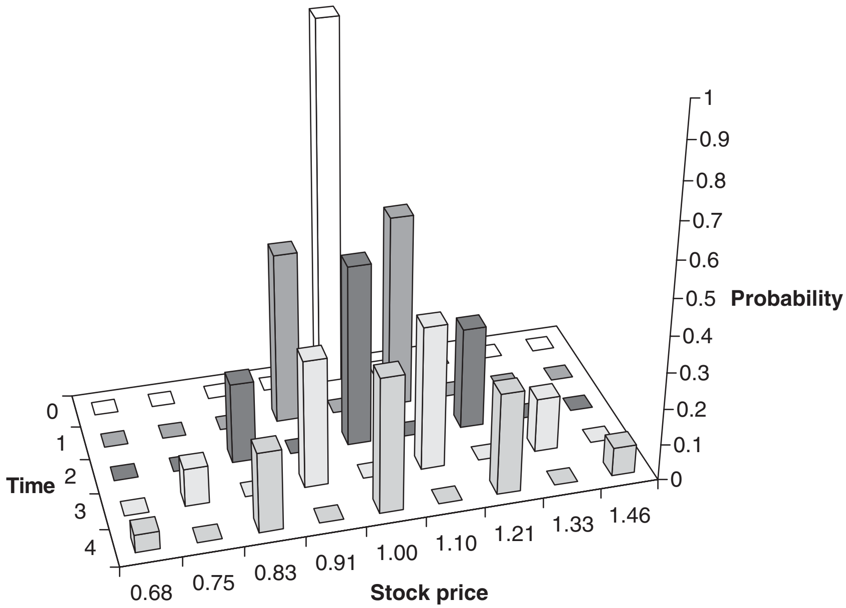


Figure 3.24 The probability distribution of future asset prices.

A few steps in a tree

3.17 VALUING BACK DOWN THE TREE

We certainly know V^+ and V^- at expiry, time T, because we know the option value as a function of the asset; this is the payoff function.

If we know the value of the option at expiry we can find the option value at the time $T - \delta t$ for all values of S on the tree. But knowing these values means that we can find the option values one step further back in time.

- Thus we work our way back down the tree until we get to the root.

This root is the current time and asset value, and thus we find the option value today.

This algorithm is shown schematically over the next few pages. In this example I have used the choices of u, v and p' described in the appendix to this chapter with $S = 100$, $\delta t = 1/12$, $r = 0.1$, and $\sigma = 0.2$. The option is a European call with a strike of 100 and four months to expiration.

Using these numbers we have $u = 1.0604$, $v = 0.9431$ and $p' = 0.5567$. As an example, after one time step the asset takes either the value $100 \times 1.0604 = 106.04$ or $100 \times .9431 = 94.31$. Working back from expiry, the option value at the time step before expiry when $S = 119.22$ is given by

$$e^{-0.1\times 0.0833}(0.5567 \times 26.42 + (1 - 0.5567) \times 12.44) = 20.05.$$

Working right back down the tree to the present time, the option value when the asset is 100 is 6.13.

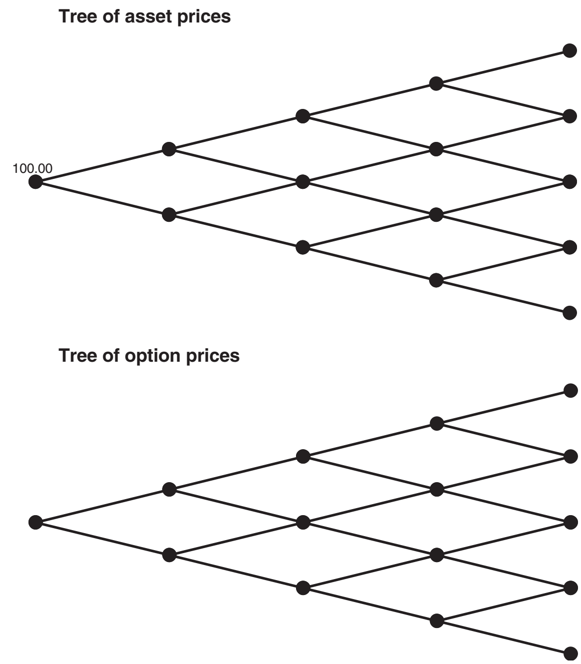


Figure 3.25 The two trees, asset and option.

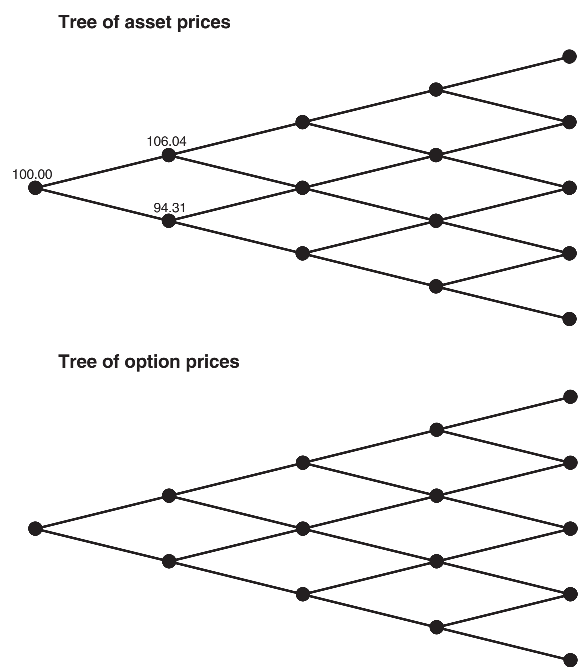


Figure 3.26 Start building up the stock-price tree.

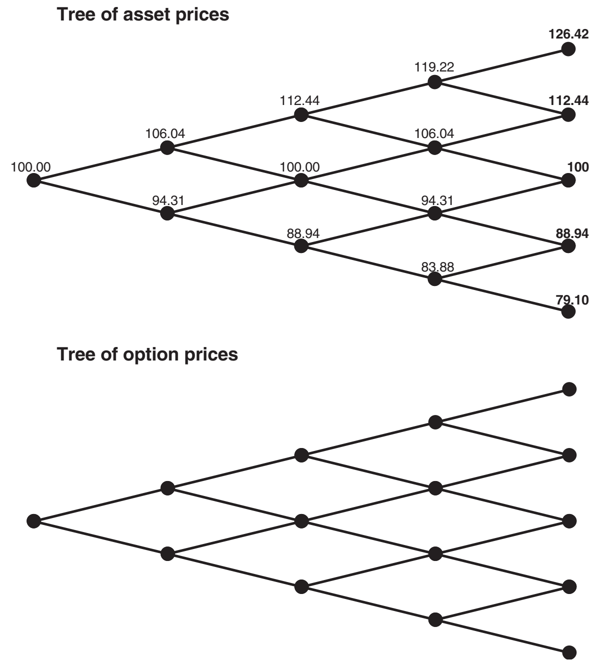


Figure 3.27 The finished stock tree.

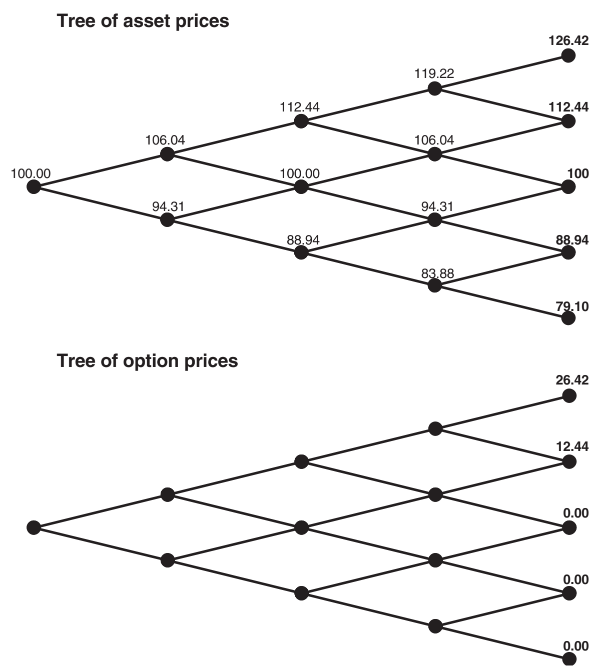


Figure 3.28 The option payoff.

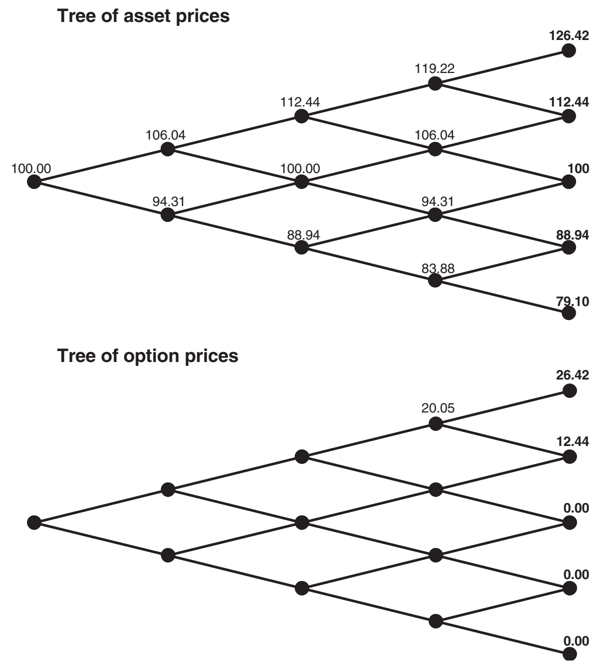


Figure 3.29 Work backwards one 'node' at a time.

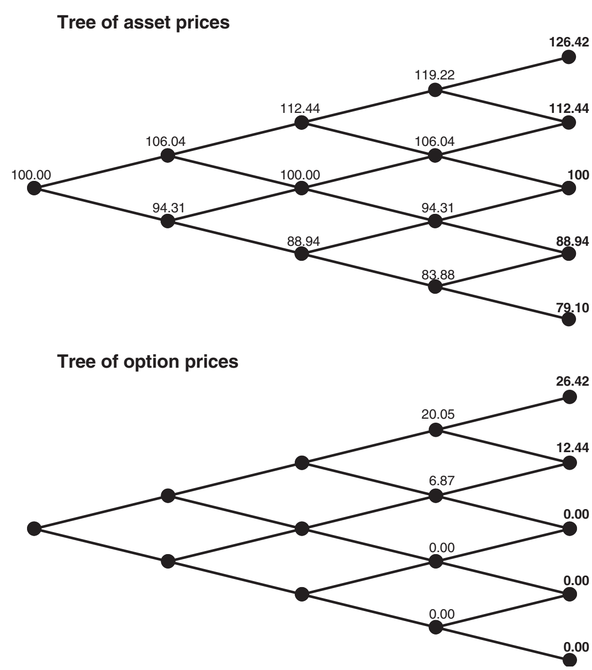


Figure 3.30 First time step completed.

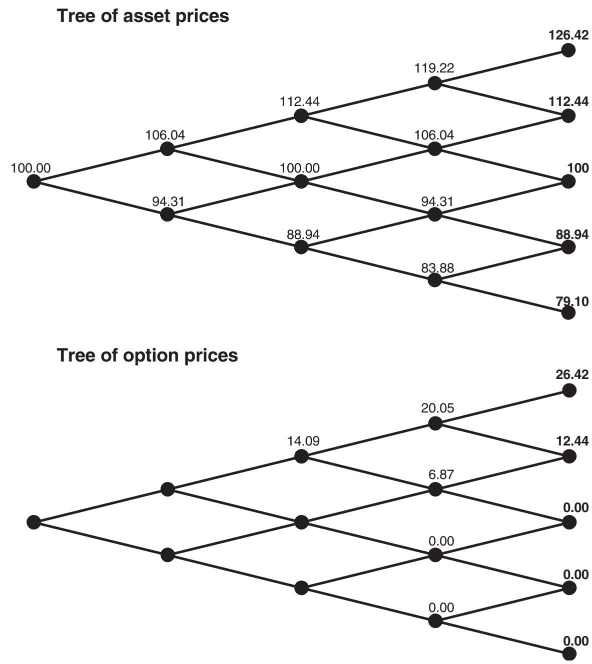


Figure 3.31 Starting on next time step.

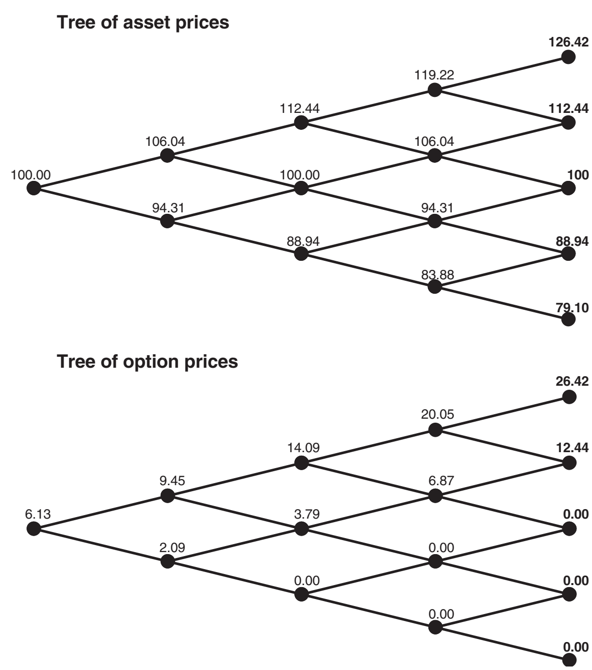
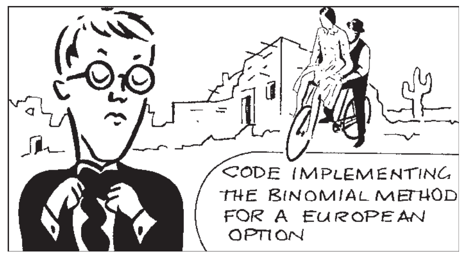


Figure 3.32 The finished option-price tree. Today's option price is therefore 6.13.

3.18 PROGRAMMING THE BINOMIAL METHOD

In practice, the binomial method is programmed rather than done on a spreadsheet. Here is a function that takes inputs for the underlying and the option, using an externally defined payoff function.[2] Key points to note about this program concern the building up of the arrays for the asset `S()` and the option `V()`. First of all, the asset array is built up only in order to find the final values of the asset at each node at the final time step, expiry. The asset values on other nodes are never used. Second, the argument `j` refers to how far up the asset is from the lowest node *at that time step*.




```
Function Price(Asset As Double, Volatility _
                        As Double, IntRate _
                        As Double, Strike _
                        As Double, Expiry _
                        As Double, NoSteps _
                        As Integer)
ReDim S(0 To NoSteps)
ReDim V(0 To NoSteps)
time step = Expiry / NoSteps
DiscountFactor = Exp(-IntRate * time step)
temp1 = Exp((IntRate + Volatility * Volatility) _
                                          * time step)
temp2 = 0.5 * (DiscountFactor + temp1)
u = temp2 + Sqr(temp2 * temp2 - 1)
d = 1 / u
p = (Exp(IntRate * time step) - d) / (u - d)

S(0) = Asset
For n = 1 To NoSteps
    For j = n To 1 Step -1
        S(j) = u * S(j - 1)
    Next j
        S(0) = d * S(0)
Next n

For j = 0 To NoSteps
    V(j) = Payoff(S(j), Strike)
Next j

For n = NoSteps To 1 Step -1
    For j = 0 To n - 1
        V(j) = (p * V(j + 1) + (1 - p) * V(j)) _
            * DiscountFactor
    Next j
Next n
Price = V(0)
End Function
```

Working code

[2] This parameterization of the binomial method is the one explained in the appendix to this chapter.

Here is the externally defined payoff function `Payoff(S, Strike)` for a call.

```
Function Payoff(S, K)
Payoff = 0
If S > K Then Payoff = S - K
End Function
```

Because I never use the asset nodes other than at expiry I could have used only the one array in the above, with the same array being used for both S and V. I have kept them separate to make the program more transparent. Also, I could have saved the values of V at all of the nodes; in the above I have only saved the node at the present time. Saving all the values will be important if you want to see how the option value changes with the asset price and time, if you want to calculate greeks for example.

In Figure 3.33 I show a plot of the calculated option price against the number of time steps using this algorithm. The inset figure is a close-up. Observe the oscillation. In this example, an odd number of time steps gives an answer that is too high and an even an answer that is too low.

3.19 THE GREEKS

The greeks are defined as derivatives of the option value with respect to various variables and parameters. These greeks will later be very important when we talk about risk management. It is important to distinguish whether the differentiation is with respect to a variable or a parameter (it could, of course, be with respect to both). If the differentiation is only with respect to the asset price and/or time then there is sufficient information in our binomial tree to estimate the derivative. It may not be an accurate estimate, but it will be *an* estimate. The option's delta, gamma and theta, defined below can all be estimated from the tree.

On the other hand, if you want to examine the sensitivity of the option with respect to one of the parameters, then you must perform another binomial calculation.

Let me take these two cases in turn.

From the binomial model the option's delta is defined by

$$\frac{V^+ - V^-}{(u - v)S}.$$

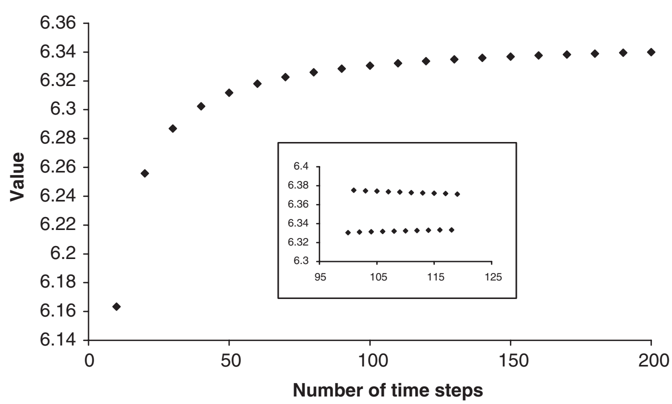


Figure 3.33 Option price as a function of number of time steps.

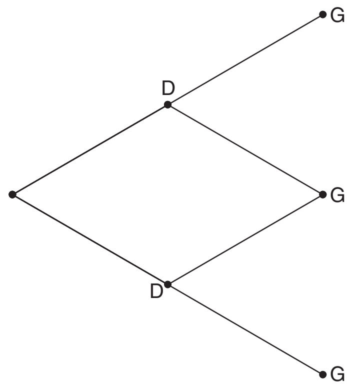


Figure 3.34 Calculating the delta and gamma.

We can calculate this quantity directly from the tree. Referring to Figure 3.34, the delta uses the option value at the two points marked 'D,' together with today's asset price and the parameters u and v. This is a simple calculation.

In the limit as the time step approaches zero, the delta becomes

$$\frac{\partial V}{\partial S}.$$

The gamma of the option is also defined as a derivative of the option with respect to the underlying, the sensitivity of the delta to the asset.

$$\frac{\partial^2 V}{\partial S^2}.$$

To estimate this quantity using our tree is not so clear. (It will be much easier when we use a finite-difference grid.) However, gamma, being the sensitivity of the delta to the underlying, is a measure of how much we must rehedge at the next time step. But we can calculate the delta at points marked with a D in Figure 3.34 from the option value one time step further in the future. The gamma is then just the change in the delta from one of these to the other divided by the distance between them. This calculation uses the points marked 'G' in Figure 3.34.

The theta of the option is the sensitivity of the option price to time, assuming that the asset price does not change. (Again, this is easier to calculate from a finite-difference grid.) An obvious choice for the discrete-time definition of theta is to interpolate between V^+ and V^- to find a theoretical option value *had the asset not changed* and use this to estimate

$$\frac{\partial V}{\partial t}.$$

This results in

$$\frac{\frac{1}{2}(V^+ + V^-) - V}{\delta t}.$$

As the time step gets smaller and smaller these greeks approach the Black–Scholes continuous-time values, which we'll be seeing shortly.

Estimating the other type of greeks, the ones involving differentiation with respect to parameters, is slightly harder. They are harder to calculate in the sense that you must perform a second binomial calculation. I will illustrate this with the calculation of the option's vega.

The vega is the sensitivity of the option value to the volatility

$$\frac{\partial V}{\partial \sigma}.$$

Suppose we want to find the option value and vega when the volatility is 20%. The most efficient way to do this is to calculate the option price twice, using a binomial tree, with two different values of σ. Calculate the option value using a volatility of $\sigma \pm \epsilon$, for a small number ϵ; call the values you find $V_\pm$. The option value is approximated by the average value

$$V = \tfrac{1}{2}(V_+ + V_-)$$

and the vega is approximated by

$$\frac{V_+ - V_-}{2\epsilon}.$$

The importance of these greeks in risk management will become increasingly apparent as you read this book.

3.20 EARLY EXERCISE

American-style exercise is easy to implement in a binomial setting. The algorithm is identical to that for European exercise with one exception. We use the same binomial tree, with the same u, v and p, but there is a slight difference in the formula for V. We must ensure that there are no arbitrage opportunities at any of the nodes.

For reasons which will become apparent, I'm going to change my notation now, making it more complex but more informative. Introduce the notation S_j^n to mean the asset price at the nth time step, at the node j from the bottom, $0 \le j \le n$. This notation is consistent with the code above. In our lognormal world we have

$$S_j^n = S u^j v^{n-j},$$

where S is the current asset price. Also introduce V_j^n as the option value at the same node. Our ultimate goal is to find V_0^0 knowing the payoff, i.e. knowing V_j^M for all $0 \le j \le M$ where M is the number of time steps.

Returning to the American option problem, arbitrage is possible if the option value goes below the payoff at any time. If our theoretical value falls below the payoff then it is time to exercise. If we do then exercise the option its value and the payoff must be the same. If we find that

$$\frac{V_{j+1}^{n+1} - V_j^{n+1}}{u - v} + \frac{1}{1 + r\,\delta t}\frac{uV_j^{n+1} - vV_{j+1}^{n+1}}{u - v} \ge \text{Payoff}(S_j^n)$$

then we use this as our new value. But if

$$\frac{V_{j+1}^{n+1} - V_j^{n+1}}{u - v} + \frac{1}{1 + r\,\delta t}\frac{uV_j^{n+1} - vV_{j+1}^{n+1}}{u - v} < \text{Payoff}(S_j^n)$$

we should exercise, giving us a better value of

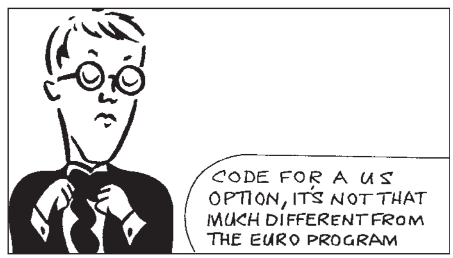
$$V_j^n = \text{Payoff}(S_j^n).$$

We can put these two together to get

$$V_j^n = \max\left(\frac{V_{j+1}^{n+1} - V_j^{n+1}}{u - v} + \frac{1}{1 + r\,\delta t}\frac{uV_j^{n+1} - vV_{j+1}^{n+1}}{u - v}, \text{Payoff}(S_j^n)\right)$$

instead of (3.3). This ensures that there are no arbitrage opportunities. This modification is easy to code, but note that the payoff is a function of the asset price at the node in question. This is new and not seen in the European problem for which we did not have to keep track of the asset values on each of the nodes.

Below is a function for calculating the value of an American-style option. Note the differences between this program and the one for the European-style exercise. The code is the same except that we keep track of more information and the line that updates the option value incorporates the no-arbitrage condition.




```
Function USPrice(Asset As Double, Volatility _
                       As Double, IntRate As _
                       Double, Strike As _
                       Double, Expiry As _
                       Double, NoSteps _
                       As Integer)
ReDim S(0 To NoSteps, 0 To NoSteps)
ReDim V(0 To NoSteps, 0 To NoSteps)
time step = Expiry / NoSteps
DiscountFactor = Exp(-IntRate * time step)
temp1 = Exp((IntRate + Volatility * Volatility) * time step)
temp2 = 0.5 * (DiscountFactor + temp1)
u = temp2 + Sqr(temp2 * temp2 - 1)
d = 1 / u
p = (Exp(IntRate * time step) - d) / (u - d)

S(0, 0) = Asset
For n = 1 To NoSteps
    For j = n To 1 Step -1
        S(j, n) = u * S(j - 1, n - 1)
    Next j
        S(0, n) = d * S(0, n - 1)
Next n

For j = 0 To NoSteps
    V(j, NoSteps) = Payoff(S(j, NoSteps), Strike)
Next j
For n = NoSteps To 1 Step -1
    For j = 0 To NoSteps - 1
        V(j, n - 1) = max((p * V(j + 1, n) + (1 - p) * V(j, n)) _
            * DiscountFactor, Payoff(S(j, n - 1), Strike))
    Next j

Next n
USPrice = V(0, 0)
End Function
```

Working code

3.21 THE CONTINUOUS-TIME LIMIT

As I mentioned, the binomial model is useful for explaining delta hedging and risk neutrality but not so great as a numerical method. But it can also lead us to the famous Black–Scholes equation. The binomial model is a discrete-time model whereas Black–Scholes is in continuous time. So let's examine (3.3) as $\delta t \to 0$.

First of all, we have chosen

$$u \sim 1 + \sigma\sqrt{\delta t}$$

and

$$v \sim 1 - \sigma\sqrt{\delta t}.$$

Next we write

$$V = V(S, t), \quad V^+ = V(uS, t + \delta t) \quad \text{and} \quad V^- = V(vS, t + \delta t).$$

Expanding these expressions in Taylor series[3] for small δt and substituting into (3.2) we find that

$$\Delta \sim \frac{\partial V}{\partial S} \quad \text{as} \quad \delta t \to 0.$$

Thus the binomial delta becomes, in the limit, the Black–Scholes delta.

Similarly, we can substitute the expressions for V, V^+ and V^- into (3.3) to find

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

This is the Black–Scholes equation. Again, the drift rate μ has disappeared from the equation. We'll be seeing a better derivation of this equation soon.

This famous equation will be derived later using stochastic calculus rather than via the binomial model. The stochastic calculus derivation is far more useful than the binomial, being far easier to generalize. And from the next chapter on there will be no more mention of u, v and p and the binomial model, everything will be in terms of drift rate and volatility (and even the drift rate disappears from most basic models, so it's really only volatility we'll be talking about).

3.22 SUMMARY

In this chapter I described the basics of the binomial model, deriving pricing equations and algorithms for both European- and American-style exercises. The method can be extended in many ways, to incorporate dividends, to allow Bermudan exercise, to value path-dependent contracts and to price contracts depending on other stochastic variables such as interest rates. I have not gone into the method in any detail for the simple reason that the binomial method is just a simple version of an explicit finite-difference scheme. As such it will be discussed in depth in Chapter 28. Finite-difference methods have an obvious advantage over the binomial method; they are far more flexible. I've said it a million times already, it seems, but let me say one more time: The binomial model is great for getting intuition about delta hedging and risk-neutral pricing but not for real-life pricing. So we won't be seeing any more of the binomial model from Chapter 4 on.

---

[3] If you are rusty on Taylor series then Appendix A contains a useful recap.

FURTHER READING

- The original binomial concept is due to Cox. *et al*. (1979).
- Almost every book on options describes the binomial method in more depth than I do. One of the best is Hull (2005) who also describes its use in the fixed-income world.

WELCOME TO MY WORLD

I'd like to introduce you to the inhabitants of planet Risk Neutral, in a distant, imaginary part of the universe. The people of Risk Neutral are strange, ethereal creatures, with little care for the pleasures of the flesh. Satisfaction to them comes from the world of the mind, the world of ideas, concepts, symbols, and especially abstract probability theory. Yes, they're very much like UK academics.

Risk Neutrals do not care for money. Their culture has no concept of 'risk.' Since they have no concept of risk, they also do not have any need to be compensated for taking risk. To them value and expectation are identical.

If we were to send a delegation of Earthlings to planet Risk Neutral with our binomial model and option pricing question they might behave as follows.

'Gentlemen of Risk Neutral,' we say, 'please would you put your undoubted mathematical skills to use on our perplexing problem from the world of finance?'

'We can but try,' they reply.

'We have a stock that is currently valued at \$100. Tomorrow it will be worth either \$101 or \$99. The probability of the stock being at \$101 is...' We are just about to say '60%,' the result of many months of painstaking statistical analysis. But before we can finish our sentence we are interrupted.

'Do not tell us the probability. We are perfectly capable of working this out for ourselves.'

How is this possible? How can the Risk Neutrals calculate this probability without any more data than the \$100, the \$101 and the \$99?

Recall that Risk Neutrals do not understand risk, that they ought to have a positive expectation for a risky investment. Therefore they reckon that the \$100 must be the expected stock price tomorrow, suitably discounted. (They do have interest-bearing bank accounts on Risk Neutral. But at the time of our visit interest rates were conveniently zero.)

'What is the probability then?' we ask.

'Clearly it is 50%.'

If interest rates are zero then they have solved the simple problem

$$p'101 + (1 - p')99 = 100.$$

The p' is therefore 0.5.

Now this is wrong. The probability is not 0.5, it *is* 0.6. However, we Earthlings are far too polite to disabuse them. After all we have only just met. So we continue with our problem. We explain to them about the idea of options and payoffs.

'The option payoff is one dollar if the stock rises and zero otherwise, what is the value of the option today?'

'It must be zero point five multiplied by one, plus zero point five multiplied by zero. So zero point five.' Their answer is 0.5. They have calculated a simple expectation based on their incorrect probability.

Yet this is correct. They have found the correct answer despite making a fundamental mistake; they have valued in terms of expectations. But this mistake has been made twice. Once they have calculated probability from price, and the second time price from probability. The second mistake reverses the first.

This method pricing always works. If interest rates were not zero on Risk Neutral then they would need to present value twice in the above calculations, in both the stock expectations calculation that yielded the risk-neutral probability p' and in the option valuation calculation. On the Risk Neutral planet they believe, erroneously, that all traded investments grow on average at the risk-free interest rate.

As we are leaving the planet to return to Earth we overhear two Risk Neutrals talking about trying their hand[4] at investing in the options market. They are in for a shock. They are very rapidly going to learn about risk. They may have successfully priced the option, but they have yet to discover hedging.

Time Out...

Risk neutrality again

My experience teaching quantitative finance at all levels is that risk neutrality is a very hard concept to grasp. So let's take another look at it.

- Hedging is used to eliminate risk.
- In simple models, hedging can be used to eliminate all risk from an option position.
- As well as eliminating risk, hedging removes dependence of an option value on the direction of an asset.
- If we don't care whether the asset price rises or falls, we shouldn't care about the probability of the rise or fall.
- The risk-neutral random walk is one that has the same volatility as the real asset random walk but a drift rate that is the same as the risk-free interest rate and not the real drift rate
- The punchline is that the option value is the present value of the option payoff under a risk-neutral random walk.

APPENDIX: ANOTHER PARAMETERIZATION

The three constants u, v and p are chosen to give the binomial walk the same drift and standard deviation as the stock we are trying to model. Having only these two equations for the three parameters gives us one degree of freedom in this choice. I've chosen the parametrization that results in the tidiest math. But this degree of freedom is often used

[4] Actually it's more like a paw.

to give the random walk the further property that after an up and a down movement (or a down followed by an up) the asset returns to its starting value, S.[5] This gives us the requirement that

$$v(uS) = u(vS) = S$$

i.e.

$$uv = 1. \tag{3.5}$$

For the binomial random walk to have the correct drift over a time period of δt we need

$$puS + (1-p)vS = SE\left[e^{(\mu-\frac{1}{2}\sigma^2)\delta t+\sigma\phi\sqrt{\delta t}}\right] = Se^{\mu\,\delta t},$$

i.e.

$$pu + (1-p)v = e^{\mu\,\delta t}.$$

Where all these exponentials and things come from will be clear after the next chapter, so you may want to read this appendix again later. Rearranging this equation we get

$$p = \frac{e^{\mu\,\delta t} - v}{u - v}. \tag{3.6}$$

Then for the binomial random walk to have the correct variance we need (details omitted)

$$pu^2 + (1-p)v^2 = e^{(2\mu+\sigma^2)\delta t}. \tag{3.7}$$

Equations (3.5), (3.6) and (3.7) can be solved to give

$$u = \tfrac{1}{2}\left(e^{-\mu\,\delta t} + e^{(\mu+\sigma^2)\delta t}\right) + \tfrac{1}{2}\sqrt{\left(e^{-\mu\,\delta t} + e^{(\mu+\sigma^2)\delta t}\right)^2 - 4}.$$

Approximations that are good enough for most purposes are

$$u \approx 1 + \sigma\ \delta t^{1/2} + \tfrac{1}{2}\sigma^2\delta t,$$
$$v \approx 1 - \sigma\ \delta t^{1/2} + \tfrac{1}{2}\sigma^2\delta t$$

and

$$p \approx \frac{1}{2} + \frac{(\mu - \frac{1}{2}\sigma^2)\delta t^{1/2}}{2\sigma}.$$

Of course, if this is being used for pricing options you must replace the μ with r everywhere.

[5] Other choices are possible. For example, sometimes the probability of an up move is set equal to the probability of a down move, i.e. $p = 1/2$.

EXERCISES

1. Solve the three equations for u, v and p using the alternative condition $p = \frac{1}{2}$ instead of the condition that the tree returns to where it started, i.e. $uv = 1$.
2. Starting from the approximations for u and v, check that in the limit $\delta t \to 0$ we recover the Black–Scholes equation.
3. A share price is currently \$80. At the end of three months, it will be either \$84 or \$76. Ignoring interest rates, calculate the value of a three-month European call option with exercise price \$79. You must use both the methods of setting up the delta-hedged portfolio, and the risk-neutral probability method.
4. A share price is currently \$92. At the end of one year, it will be either \$86 or \$98. Calculate the value of a one-year European call option with exercise price \$90 using a single-step binomial tree. The risk-free interest rate is 2% p.a. with continuous compounding. You must use both the methods of setting up the delta-hedged portfolio, and the risk-neutral probability method.
5. A share price is currently \$45. At the end of each of the next two months, it will change by going up \$2 or going down \$2. Calculate the value of a two month European call option with exercise price \$44. The risk-free interest rate is 6% p.a. with continuous compounding. You must use both the methods of setting up the delta-hedged portfolio, and the risk-neutral probability method.
6. A share price is currently \$63. At the end of each three month period, it will change by going up \$3 or going down \$3. Calculate the value of a six month American put option with exercise price \$61. The risk-free interest rate is 4% p.a. with continuous compounding. You must use both the methods of setting up the delta-hedged portfolio, and the risk-neutral probability method.
7. A share price is currently \$15. At the end of three months, it will be either \$13 or \$17. Ignoring interest rates, calculate the value of a three-month European option with payoff $\max(S^2 - 159, 0)$, where S is the share price at the end of three months. You must use both the methods of setting up the delta-hedged portfolio, and the risk-neutral probability method.
8. A share price is currently \$180. At the end of one year, it will be either \$203 or \$152. The risk-free interest rate is 3% p.a. with continuous compounding. Consider an American put on this underlying. Find the exercise price for which holding the option for the year is equivalent to exercising immediately. This is the break-even exercise price. What effect would a decrease in the interest rate have on this break-even price?
9. A share price is currently \$75. At the end of three months, it will be either \$59 or \$92. What is the risk-neutral probability that the share price increases? The risk-free interest rate is 4% p.a. with continuous compounding.